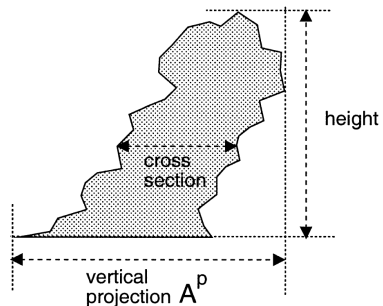


Large-eddy simulations (LES) of shallow cumulus. Each cloud gets one size; the size histogram shows which clouds dominate cloud cover and vertical transport.



$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

Three LES cases

BOMEX: ocean, trade winds.

SCMS: land, Florida.

ARM: land, diurnal cycle.

Symbols

cloud connected set of saturated grid cells

A^p vertically projected area of one cloud: its footprint seen from above

l cloud size, m

$\mathcal{N}(l)$ number of clouds per unit size, m^{-1}

N total number of clouds in the domain

Neggers et al. (2003): cloud-size statistics

Size Statistics of Cumulus Cloud Populations in Large-Eddy Simulations, Neggers, Jonker and Siebesma (2003). The study measures the sizes of shallow cumulus clouds in LES and their contributions to cloud cover and vertical transport. BOMEX represents steady trade-wind cumulus over the ocean. SCMS and ARM represent continental shallow cumulus, with a diurnal cycle in ARM. The reference horizontal grid spacing is 50 m, in a 6.4 km domain. Repeated snapshots sample tens of thousands of clouds per case.

A cloud is a connected set of saturated grid cells. Figure 3 shows the distinction between a cross-section at one height and the vertical projection of the whole cloud. The chosen size is $l = \sqrt{A^p}$, the side of a square with the cloud's projected area, not a radius or a diameter. It is chosen so that LES can be compared with satellite images, which see only projections. The size density $\mathcal{N}(l)$ gives counts per unit size, and its integral is the total count. The bottom line summarizes the actual measurement procedure. Source: Sections 2–3 and Appendix, Figure 3, Equations (1)–(2).

Find each cloud in 3D → project it vertically → histogram of the sizes l

Small clouds are by far the most numerous. Below a size ℓ_d their number falls as a power law of size; above ℓ_d it falls faster.

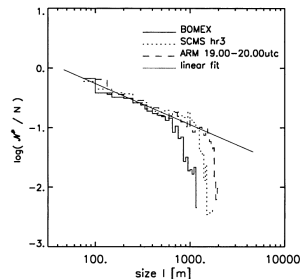


Fig. 4a: size density against l .

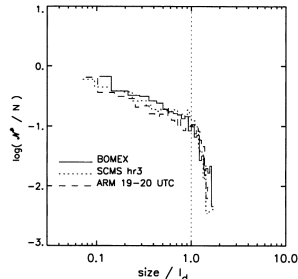


Fig. 6: against l/ℓ_d , the cases collapse.

$$\mathcal{N}(l) = al^b$$

$$b = -1.70 \text{ for } l < \ell_d$$

Symbols

ℓ_d scale break: size where the exponent changes. BOMEX 700 m, SCMS 1050 m, ARM 400 $\rightarrow$ 1250 m in the afternoon

a, b fit constants; b is the exponent
 $\mathcal{N}^* = l \ln(10) \mathcal{N}$, clouds per decade of size (plotted), slope $b + 1 = -0.70$

A power law below a variable size break

Figure 4a shows the normalized size density in logarithmic bins. Below the scale break, the three cases have approximately the same slope. The fitted log-bin slope is -0.70 , corresponding to the exponent $b = -1.70$ of $\mathcal{N}(l) = al^b$ because $\mathcal{N}^* = l \ln(10) \mathcal{N}$: a power law per decade has its exponent raised by one. The supported range spans roughly one decade between the grid scale and the break.

Above ℓ_d , the density decreases faster. Table 2 gives 700 m for BOMEX and 1050 m for SCMS. During the ARM afternoon, the break increases from 400 to 1250 m. Increasing the number of sampled clouds does not remove the break.

Figure 6 plots the three size densities against l/ℓ_d . Their approximate collapse supports a common functional form for these simulated cumulus populations, with a case-dependent break: the break is the only variable. Its physical origin remains unresolved; the paper mentions subcloud-layer depth and humidity fluctuations as candidates. It is a cloud-size scale, not an inter-cloud distance. Source: Section 4a, Figures 4a and 6, Table 2, Equations (8)–(9).

Same exponent in all cases: only the break ℓ_d changes

Weighting every cloud by its area gives the cloud cover contributed by each size. It grows up to ℓ_d and drops beyond, so mid-size clouds cover most of the sky.

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y} \propto l^{b+2} = l^{0.30} \quad (l < \ell_d)$$

$$a^p = \int_0^\infty \alpha^p(l) dl$$

Symbols

$\alpha^p(l)$ cloud cover from clouds of size l , per unit size, m^{-1}

a^p projected cloud cover: fraction of the domain covered, seen from above

$L_x L_y$ horizontal area of the domain

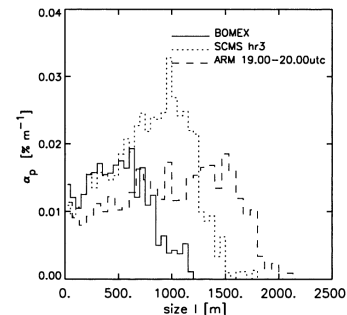


Fig. 7a: $\alpha^p(l)$, three cases.

The size break sets the dominant cloud cover

Figure 7a gives the contribution to projected cloud cover per unit size. Each cloud contributes its projected area l^2 , so $\alpha^p(l) = l^2 \mathcal{N}(l) / (L_x L_y)$, where $L_x L_y$ is the horizontal domain area. For a power law with exponent b , the cover decomposition scales as l^{b+2} . Below the break, $b = -1.70$ gives a positive exponent, 0.30: the many small clouds are too small to dominate. Above the break the density declines faster than l^{-2} , so the cover decomposition decreases. The peak therefore lies at an intermediate size near the break. Without a break, an exponent above -2 would let the largest clouds dominate.

This is a decomposition per unit linear size, not per logarithmic bin. The total projected cover is $a^p = \int_0^\infty \alpha^p(l) dl$. The paper notes that real shallow cumulus fields also show this intermediate dominant size. Source: Section 3, Section 4b, Figure 7a, Equations (3)–(4) and (10).

Cloud cover peaks near the break ℓ_d

Mass flux is cloud cover times vertical velocity. Larger clouds rise faster, so transport is dominated by somewhat larger clouds; the smallest carry almost nothing.

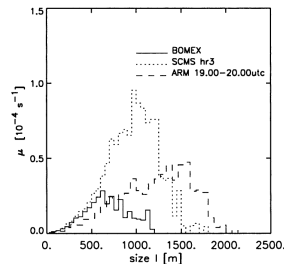
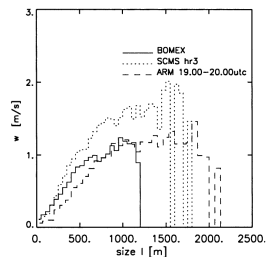


Fig. 7b: mean vertical velocity w .

Fig. 7c: mass flux $\mu(l)$.

$$\mu(l, z) = \alpha(l, z) w(l, z)$$

$$\mu(l) = \frac{1}{h_c} \int_{h_c} \mu(l, z) dz$$

Symbols

$w(l, z)$ mean vertical velocity of clouds of size l at height z , m s^{-1}

$\alpha(l, z)$ their cloud cover at height z , per unit size

h_c depth of the cloud layer

μ mass flux divided by air density, per unit size, s^{-1}

Mass flux peaks at larger sizes than cover

Figure 7b shows that cloud-mean vertical velocity generally increases with cloud size. Figure 7c shows the resulting mass-flux decomposition. At height z , $\mu(l, z) = \alpha(l, z)w(l, z)$, where $\alpha(l, z)$ is the cloud-fraction contribution per unit size at that height. Averaging over the cloud-layer depth h_c gives $\mu(l)$. Its integral over size is the total kinematic mass flux, in m/s under this paper's convention: mass flux divided by air density.

Cloud size is always the projected size, even when evaluating the flux at a specific height, so each cloud stays in one size bin at all heights. Since w increases with size, the mass-flux maximum occurs at somewhat larger cloud sizes than the projected-cover maximum. The smallest clouds contribute very little to total vertical transport, mainly because of their very low vertical velocities.

The intermediate dominant size persists for tested grid spacings of 25–100 m. A 3.2 km domain truncates the largest clouds. Increased shear broadens projected sizes while leaving total mass flux almost unchanged. These sensitivity tests support the main transport result but do not identify the mechanism of the size break. Source: Section 3, Section 4b–e, Figure 7b–c and Figures 9–11.

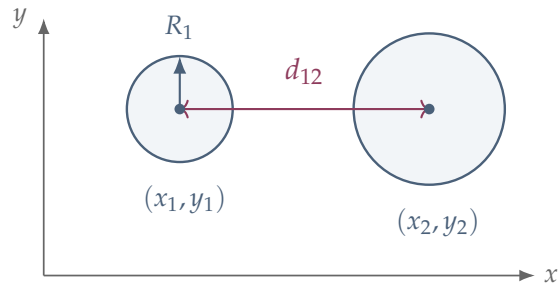
Mass flux peaks at larger cloud sizes than projected cover

No 2D distribution of thermals

5

Our question: how are thermals arranged in a horizontal plane? The paper measures clouds seen from above, a different object.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why cloud sizes do not answer it

A cloud is saturated air; a thermal can stay dry and never form a cloud.

Projection merges all heights and widens with tilt and shear.

Projected cumulus size does not determine thermal size or spacing

Scope: §3–5 and Appendix. The sketch defines the quantities sought. It represents no observed or simulated thermal field.

Paper review: Neggers et al. (2003) | Slide 5

No 2D distribution of thermals

Our reading sought distributions of thermals in a horizontal plane: their size at a fixed height, their positions and their separations. The diagram only defines these three quantities. Its circles, radii and separation are schematic, not data or a spatial model.

Neggers et al. report projected cloud sizes and cloud transport statistics. A cloud is a saturated object, whereas a thermal can be unsaturated and can feed a cloud without sharing its projected footprint. Projection also combines geometry over height and responds to tilt and shear. The paper does not establish a cloud-to-thermal size mapping and does not report thermal centres or inter-thermal distances. The exponent -1.70 and the break ℓ_d therefore remain results about the measured cumulus population. Source: Sections 3–5 and Appendix.